

Question number	Answer	Additional guidance	Mark
2(a)(i)	D sensory neurone is correct A is not correct because a motor neurone does not continue a reflex action after the receptor. B is not correct because the relay neurone does not continue a reflex action after the receptor. C is not correct because a Schwann cell does not continue a reflex action after the receptor.		(1)

Question number	Answer	Additional guidance	Mark
2(a)(ii)	C is the correct answer receptor → spinal cord → muscle B is not correct because muscle → receptor → brain is not the correct pathway C is not correct because muscle → spinal cord → brain is not the correct pathway D is not correct because spinal cord → brain → Muscle is not the correct pathway.		(1)

Question number	Answer	Additional guidance	Mark
2(b)(i)	A description that includes three of the following points: • correct description of sodium potassium pump (1) • the sodium ion channels are closed (1) • the membrane is relatively permeable to potassium ions (1) • at rest high potassium ions (K^+) inside the axon, high sodium ions (Na^+) outside axon (1)		(3)

Question number	Answer	Additional guidance	Mark
2(b)(ii)	An answer that includes the following points: • pressure deforms cell membranes in the Pacinian corpuscle (1) • an influx of sodium ions into the axon (1) • details of the action potential in the axon (1)	If they refer to opening the sodium channels it could be an alternative to second point	(3)

Question number	Answer	Additional guidance	Mark
2(b)(iii)	An answer that includes three of the following points: • the threshold has been reached (1) • results in action potential (1) • action potentials are all-or-nothing, since either they occur fully or they do not occur at all. (1)		(3)

